

Assignment 1
Topics in Statistics: MATH 598-03/782-03
Due date: Friday, October 02, 2026, at 11:59pm
To be submitted online via myCourses

Question 1. The data in this question concern the relationship between the level of prostate-specific antigen (PSA) and several clinical measures for individuals who were about to undergo a radical prostatectomy. The dataset contains 97 observations on the PSA response variable and the following clinical measures (covariates):

x_1 . log cancer volume (lcavol)

x_2 . log prostate weight (lweight)

x_3 . age

x_4 . log of the amount of benign prostatic hyperplasia (lbph)

x_5 . seminal vesicle invasion (svi)

x_6 . log of capsular penetration (lcp)

x_7 . Gleason score (gleason)

x_8 . percent of Gleason scores 4 or 5 (pgg45).

The data are posted on **myCourses**. Standardize each of the covariates $x_1, \dots, x_8$ so that they have mean zero and unit variance, and denote the resulting standardized covariates by $x_1^*, \dots, x_8^*$. The response variable in our analysis is the logarithm of the PSA level, defined as $y = \log(\text{psa})$. Consider a multiple linear regression model

$$y_i = \beta_0 + \beta_1 x_1^* + \beta_2 x_2^* + \dots + \beta_8 x_8^* + \varepsilon_i \quad , \quad i = 1, 2, \dots, 97,$$

where $E(\varepsilon_i) = 0$ and $\text{Var}(\varepsilon_i) = \sigma^2$. Here $d = 8, n = 97$. Using R or RStudio, fit all the possible submodels of the above model to the data, including the model with

an intercept only, and also the full model. This corresponds to the collection $\mathcal{M}$ of all submodels discussed in class.

1. Compute sum of squares of residuals, $\text{SSR}(M)$, corresponding to each fitted linear submodel, say, M . Construct a plot of $\text{SSR}(M)$ versus the number of covariates, d_M , included in each submodel. Do you see any specific pattern in the plot? Explain your observation.
2. Use forward, backward, and their hybrid version (stepwise) selection methods to see what submodel(s) these three methods provide for this data.
3. Fix $\nu = 0.1$ for both forward stagewise linear regression (FSLR) and L_2 -Boosting. Using existing packages in **R** or **RStudio**, or by implementing the two methods directly, apply these procedures to the data to determine the final models they suggest. You may choose the number of iterations for each method.

Question 2. (A simulation study). Consider a multiple linear regression model

$$Y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \beta_3 x_{i3} + \beta_4 x_{i4} + \beta_5 x_{i5} + \varepsilon_i = \mathbf{x}_i^\top \boldsymbol{\beta} + \varepsilon_i \quad ; \quad i = 1, 2, \dots, n$$

where $\mathbf{x}_i^\top = (1, x_{i1}, x_{i2}, x_{i3}, x_{i4}, x_{i5})$, $\boldsymbol{\beta}^\top = (\beta_0, \beta_1, \beta_2, \beta_3, \beta_4, \beta_5)$, and $\varepsilon_i \sim N(0, 1)$. The five explanatory variables $(x_1, x_2, x_3, x_4, x_5)$ are generated from a multivariate normal distribution with a mean zero, variance one and the pairwise autoregressive correlation $\text{cor}(x_j, x_k) = \rho^{|j-k|}$, $j \neq k$, for $\rho = 0.0, 0.5, 0.75$. The corresponding three design matrices $\mathbf{X}_{I, n \times 6}$, $\mathbf{X}_{II, n \times 6}$ and $\mathbf{X}_{III, n \times 6}$ are posted on **myCourses**. The true value of the regression coefficients in the above model are set to

$$(\beta_0, \beta_1, \beta_2, \beta_3, \beta_4, \beta_5) = (1.5, 0.0, 2.0, 0.0, 1.5, 0.0).$$

Using the above setting, generate $R = 100$ random samples of sizes $n = 50$ from the true model, corresponding to each of the given design matrices.

We would like to compare finite sample performance of the model selection criteria: Mallows Cp, AIC, BIC, CV, GCV, and CV_d for $d = 5, 10$. There are $2^5 = 32$ possible submodels with intercepts. Based on each of the $R = 100$ random samples, corresponding to a value of ρ , tabulate the empirical probabilities of selecting a submodel by each of the above criterion. To reduce the size of your three tables, report only the empirical probabilities corresponding to the true submodel and also those submodels that include the true submodel.

Question 3. (Daily temperature data) The data for this question, which is posted on **myCourses** in a pdf file, shows the daily minimum temperatures (Y) in January averaged from 1971 through 2000; the latitudes (x_1), longitudes (x_2) and altitudes (x_3), for 25 cities in Japan. To predict the average daily minimum temperature in January, we consider the full multiple linear regression model

$$Y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \beta_3 x_{i3} + \varepsilon_i$$

where the error ε_i is assumed normally distributed with $N(0, \sigma^2)$. Due to the nature of the three explanatory variables x_1, x_2 and x_3 , it is found that fitting the above full model is not necessary, and a variable selection is needed. There are 8 possible submodels (including the model with no variables, i.e. only a model with intercept). The goal is to select the best submodel from the pool of 8 possible submodels.

Consider the variable selection criteria: Mallows Cp, AIC, BIC, CV, GCV, and CV_5 . (For the CV_5 , you may partition the data into 5 mutually exclusive subdata sets). Fit all 8 submodels and obtain the values of each criterion corresponding to each submodel. Write down the best submodel selected by each criterion. Using your results, what would be your final selected submodel for this data? Comment on your choice.

Question 4. Let Y be a real-valued continuous random variable with possible values in $\mathcal{Y} \subset \mathbb{R}$, and its true unknown probability density function is given by $g(y)$, where

$$g(y) \geq 0, \forall y \in \mathcal{Y} \quad ; \quad \int_{\mathcal{Y}} g(y) dy = 1.$$

We have proposed to use a known probability density function $f(y)$ to approximate the $g(y)$, where

$$f(y) \geq 0, \forall y \in \mathcal{Y} \quad ; \quad \int_{\mathcal{Y}} f(y) dy = 1.$$

The Kullback-Leibler distance is used to measure the closeness of the two probability density functions f and g as

$$KL(g; f) = \int_{\mathcal{Y}} g(y) \log \frac{g(y)}{f(y)} dy$$

- (a) Prove that $KL(g; f) > 0$, and furthermore $KL(g; f) = 0$ if and only if $g(y) = f(y)$ for all $y \in \mathcal{Y}$.
- (b) Assume that the true model $g(y)$ is a two-sided exponential (Laplace) distribution with

$$g(y) = \frac{1}{2} e^{-|y|} \quad -\infty < y < \infty.$$

Let the approximating model f to be the normal density function

$$f(y; \boldsymbol{\theta}) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(y-\mu)^2}{2\sigma^2}} \quad -\infty < y < \infty$$

with $\boldsymbol{\theta} = (\mu, \sigma^2)$. Calculate the KL distance of f from the true g . The distance will be a function of the parameters μ and σ , say, $KL(\mu, \sigma^2)$.

- (c) Can you find a value of (μ, σ^2) that minimizes the distance $KL(\mu, \sigma^2)$?

Question 5. Let Y be a real-valued continuous random variable with the true probability density function g . We have considered to use the normal density function

$$f(y; \boldsymbol{\theta}) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(y-\mu)^2}{2\sigma^2}} \quad , \quad -\infty < y < \infty$$

with the unknown parameters $\boldsymbol{\theta} = (\mu, \sigma^2)$, to approximate g . Let $Y_1, Y_2, \dots, Y_n$ be a random sample of observations from g .

- (a) Derive the two matrices $\mathbf{I}(\boldsymbol{\theta})$ and $\mathbf{J}(\boldsymbol{\theta})$ as discussed in the class. Comment on the difference between the two matrices.
- (b) Compute the generalized dimension

$$p^* = \text{tr}\{\mathbf{J}^{-1}(\boldsymbol{\theta}_0)\mathbf{I}(\boldsymbol{\theta}_0)\}$$

where $\boldsymbol{\theta}_0 = (\mu_0, \sigma_0^2)$, with $\mu_0 = E_g(Y)$, $\sigma_0^2 = \text{Var}_g(Y)$, is denoted as the least false parameter value as discussed in my lectures.

- (c) In practice, p^* cannot be computed directly. It is often estimated by replacing $\boldsymbol{\theta}_0$ by its maximum likelihood estimator. Using this fact, derive the TIC formula corresponding to the normal parametric family.
- (d) Derive the AIC formula corresponding to the normal parametric family.
- (e) Discuss a case where TIC and AIC are equal.